Lesson 26 Cardinality of Infinite Sets

We define the cardinality of a finite set as the number of elements in a set, but this does not work for infinite sets. The definition of **cardinality of infinite sets** was due almost solely to the genius of Georg Cantor (1845 – 1918).

Definition Sets A and B have the same ***cardinality*** if there is a bijection $f: A \to B$.

This definition gives expected results for finite sets: there are bijections if and only if A and B have the same number of elements. In particular a ***proper*** (strictly smaller) subset B of a finite set A will never have the same cardinality as A since there are no bijections from A to B. However, an infinite set may have the same cardinality as its proper subsets!

Proposition 1 $\mathbb{N}$ has the *same* cardinality as $\mathbb{Z}$.

Proof We show that $\mathbb{N}$ has the *same* cardinality as $\mathbb{Z}$ with the bijection $f: \mathbb{N} \to \mathbb{Z}$,

$$\begin{array}{ccccccccc} 1, & 2, & 3, & 4, & 5, & 6, & 7, & 8, & 9, \ldots. \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ 0, & 1, & -1, & 2, & -2, & 3, & -3, & 4, & -4, \ldots. \end{array}$$

$$f(n) = (-1)^n \left\lfloor \frac{n}{2} \right\rfloor$$

We leave it as an exercise to show formally that f is one-to-one and onto.

The cardinality of an infinite set is NOT like that of a finite set. On the one hand, since $\mathbb{N}$ is a proper subset of $\mathbb{Z}$, $\mathbb{N}$ is "smaller" than $\mathbb{Z}$. On the other hand, there is a bijection between $\mathbb{N}$ and $\mathbb{Z}$, every element of $\mathbb{Z}$ can be paired with exactly one element of $\mathbb{N}$, so it is reasonable that the two sets have the same cardinality!

The cardinality of $\mathbb{N}$ is a fundamental kind of "infinity".

Definition A set is ***countable*** if it has the same cardinality as $\mathbb{N}$.

Proposition 2 A set A is countable if and only if the elements of A can be arranged in a list or a one-dimensional array: $A = \{a_1, a_2, a_3, \ldots\}$.

Proof A is countable if and only if there is a bijection $f: \mathbb{N} \to A$, and we can let $a_n = f(n)$ for every $n \in \mathbb{N}$, and A is a 1D list: $A = \{f(1), f(2), f(3), \ldots\}$.

Proposition 2 is straightforward and useful: elements of a countable set can be arranged in a line like the natural numbers. It is clear that any infinite subset of A can also be arranged in a line (just drop some elements), hence the following:

Corollary 3 An infinite subset of a countable set is countable.

We have shown that $\mathbb{Z}$ has the same cardinality of $\mathbb{N}$, even though there are "twice" as many integers as there are positive integers, Now, there are way more rational numbers than integers, and surprise: $\mathbb{Q}$ has the same cardinality of $\mathbb{N}$!

Proposition 4 $\mathbb{Q}$ is countable.

Proof Write the set of nonnegative rational numbers $\mathbb{Q}_{\geq 0}$ as a 2-dimensional array,

	0	1	2	3	4	$\cdots$
1	$0/1$	$1/1$	$2/1$	$3/1$	$4/1$	
2	$0/2$	$1/2$	$2/2$	$3/2$	$4/2$	
3	$0/3$	$1/3$	$2/3$	$3/3$	$4/3$	
4	$0/4$	$1/4$	$2/4$	$3/4$	$4/4$	
$\vdots$						

Follow the arrows to write the 2-dimensional array as a one-dimensional array:

$$\frac{0}{1}, \frac{0}{2}, \frac{1}{1}, \frac{0}{3}, \frac{1}{2}, \frac{2}{1}, \frac{0}{4}, \frac{1}{3}, \cdots$$

So, $\mathbb{Q}_{\geq 0}$ is countable. Clearly the negative rational numbers $\mathbb{Q}_{<0}$ can also be listed in a line, and we can list $\mathbb{Q}$ in a line by taking elements alternately from $\mathbb{Q}_{\geq 0}$ and $\mathbb{Q}_{<0}$. Some fractions in have the same value ($1/1 = 2/2$ etc), but this does not change the conclusion: the set of *distinct* rational numbers is countable as it is an infinite subset of $\mathbb{Q}$ (Corollary 3).

We have shown that the infinite sets $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ all have the same cardinality. We might think that all infinite sets have the same cardinality, but that is not true! The set of real numbers $\mathbb{R}$ is at a higher level of infinity than $\mathbb{N}$!

Proposition 5 The interval $[0,1)$ is uncountable.

Proof (Diagonal Argument)

Suppose that $[0,1)$ is countable, then we can write $[0,1)$ as a one-dimensional array, $\{a_1, a_2, a_3, \ldots.\}$. Let $a_k = .a_{k1}a_{k2}a_{k3}a_{k4} \ldots$ be the decimal expansion:

$$a_1 = .a_{11}a_{12}a_{13}a_{14} \ldots$$
$$a_2 = .a_{21}a_{22}a_{23}a_{24} \ldots$$
$$a_3 = .a_{31}a_{32}a_{33}a_{34} \ldots$$
$$\vdots$$

Let $b = .b_1b_2b_3b_4 \ldots$ be a number in $[0,1)$ such that $b_k \neq a_{kk}$. As $b_k \neq a_{kk}$ for all k, b is not any a_k, contradicting the premise that $[0,1) = \{a_1, a_2, a_3, \ldots.\}$, which comes from the supposition that $[0,1)$ is countable. Thus, $[0,1)$ is not countable.

Corollary 6 $\mathbb{R}$ is uncountable.

Proof Suppose $\mathbb{R}$ is countable. Since $[0,1) \subseteq \mathbb{R}$, $[0,1)$ is countable by Corollary 3; this contradicts Proposition 5.